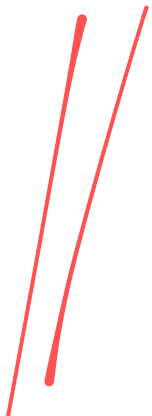
CMOS SRAM Circuit Design and Parametric Test in Nano-Scaled Technologies



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To Natasha and Daniel

To Sunanda, Aniruddh and Arushi

Foreword

Embedded SRAMs now dominate CMOS computing chips taking well over half of the total transistor count of high performance ICs. This dominance forces designers to minimize the SRAM layout area imposing a tight transistor density. This transistor circuit density presents two challenges for the test. The first is that virtually all areas of the cells are active and sensitive to particle-related defects. Secondly, parasitic coupling between cells is a major concern. This book addresses both of these problems.

The strongest approach to test method development examines the failure mechanism under test down to the transistor, interconnect and dielectric level. Test detection methods are guided by the electronic properties of the failure mechanism. This analysis style and subsequent development of a test method is called defect-based testing (DBT). This book is a strong example of the DBT thinking. The authors describe an comprehensive SRAM test that is supported by abundant simulation and silicon data. High-density embedded SRAMs require this detailed level of study to understand how to avoid the severe implications of test escapes in this critical majority region of modern ICs.

The authors also supply excellent tutorial descriptions of topics that support SRAMs. This includes design of the memory system and its components, SRAM cell stability, traditional fault models and test practices, I_{DDQ} testing, burn-in challenges and a particularly relevant Chapter 6 on soft error impact on nano-scaled SRAM cells. This book is well written and the reader is the beneficiary of the large amount of work by the authors. The book should be read and on the shelf of engineers who deal with high-performance chips.

Professor Chuck Hawkins
University of New Mexico

Preface

The process technology scaling and push for better performance enabled embedding of millions of Static Random Access Memories (SRAM) cells into contemporary ICs. In several applications, the embedded SRAMs can occupy the majority of the chip area and contain hundreds of millions of transistors. As the process technology continues to scale deeper into the nanometer region, the stability of embedded SRAM cells is a growing concern. As a consequence, large SRAM arrays impact all aspects of chip design and manufacturing because they became the yield-limiters in modern high-performance ICs. However, the robust detection of SRAM cells with marginal stability is a non-trivial task. While the traditional march tests are unable to detect unstable cells, the conventional data retention tests that are intended to expose marginal cells have unacceptable sensitivity and are uneconomical due to the long test time and high-temperature requirements. These factors show the growing disparity between the traditional SRAM test practices and the need for an economical SRAM cell stability tests that can help to achieve lower defect counts in the shipped parts. While various aspects of SRAM design and test have been addressed in special literature, no cohesive text provides a systematic overview of SRAM cell stability and the specialized design and test approaches it requires.

Therefore, the primary objective of this book is to bridge the gap between the challenges that the technology scaling brings on SRAM circuit design and the design and test solutions spanning across the process technology, circuit and system design and the testing. The book gives an overview of SRAM design and the traditional SRAM testing. It builds the reader's understanding on the detailed treatment of the aspects of SRAM cell stability and the state-of-the-art specialized stability testing techniques including the test techniques developed by the authors. In addition, since the unstable SRAM cells are more prone to soft errors, we included an overview of the sources, mechanisms and the mitigation techniques for soft errors in SRAM arrays.

The intended audience of this book is graduate students, engineers and professionals interested in developing understanding and intuition of the challenges faced by modern SRAM design and test engineers.

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Andrei Pavlov
Manoj Sachdev

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Acronyms

IC	Integrated Circuit - electronic circuit integrating active and passive components on the same semiconductor substrate
SRAM	Static Random Access Memory - semiconductor memory that is designed to hold the stored data for as long as the power is supplied without the need to refresh it
SNM	Static Noise Margin - a stability metric of an SRAM cell
SoC	System on a Chip - IC that integrates digital, analog, mixed-signal and RF functions on the same chip
ITRS	International Roadmap for Semiconductors - group of experts defining semiconductor technology roadmapping
DRT	Data Retention Test - SRAM test for data retention when an SRAM cell is powered up but not accessed
DRF	Data Retention Fault - failure to retain the data in an SRAM cell when the cell is powered up but not accessed
SF	Stability Fault - failure to retain the data in an SRAM cell in adverse operating/testing conditions
WL	Word Line - access control line of a “word” of SRAM cells
BL,BLB	Bit Line and Bit Line Bar - the true and the complementary input/output lines of a column of SRAM cells
CMOS	Complementary Metal-Oxide-Semiconductor - traditional abbreviation referring to field-effect transistors with metal (or polysilicon) gates and silicon dioxide (or high-k) gate dielectric
DPM	Defect Per Million - measure of fabrication quality of SRAM arrays
IFA	Inductive Fault Analysis - special fault analysis technique that identifies probability of a defect to occur in a given circuit layout
ECC	Error Correction Code - using redundant information to detect and correct data errors
VTC	Voltage Transfer Characteristic - output voltage of a logic gate as a function of input voltage
SER	Soft Error Rate - the rate at which a device or system encounters or is predicted to encounter soft errors. Expressed in FIT or MTBF

FIT	Failure In Time - equivalent to 1 error per billion hours of device operation
MTBF	Mean Time Between Failures - in years of device operation; 1 year MTBF is equal to ≈114,155 FIT
PVT	Process, Voltage, Temperature - a set of conditions affecting the circuit
TFT	Thin Film Transistor